



Geographical variability of environmental parameters versus GPS precision: Toward a better sampling strategy

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ABSTRACT

To characterize a sedimentary environment, it is risky to take a single sample when the spatial variability is unknown. A reference station has to reflect the natural variations in order to allow the creation of long time series. However, it can remain unclear whether the temporal changes are real or due to a spatial variation.

We highlight here the importance of spatial variability at the scale of precision of the GNSS. It appears that the number and arrangement of replicates depend on the environment and the studied parameters.

InC, TOC and TS show a sufficiently low spatial variability to allow temporal tracking using GNSS without multiplying samples. The fine fraction percent shows a high spatial variability over small distances. The study of this parameter in the framework of temporal tracking requires a knowledge of its spatial variability during each period of sampling, and hence leads to the multiplication of samples.

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1. Introduction

Most environmental studies are based on comparisons of datasets obtained at different locations usually called sampling stations. These stations are mostly selected according to a sampling scheme (statistical, geostatistical, stratified, etc.). Classically, comparisons are carried out by considering one of the sampling stations as a “reference station”. The concept of a reference station is also encountered in studies of the temporal variation of aquatic environmental quality (seasonal and inter-annual fluctuations and long-term trends). The reference station has to reflect the natural variations in the field as closely as possible to allow the creation of long time series and comparison with other stations.

Since interactions between benthic communities and local sediments give rise to rapid variations both in space and time (Birch et al., 2001; Tolhurst and Chapman, 2005), Frontier (1983) argues that it is necessary to collect replicates whatever the sampling strategy. However, this is not generally the case, as shown by Wu et al. (2008) from a review of 661 scientific articles. Hence, during most sampling campaigns, only one sample is collected at each location. This lack of replicate samples can undermine the reliability of the study. In most environmental monitoring programmes, biological and water samples are more often replicated, but such an approach is less evident for sediment sampling. This is particu-

larly problematic because sediments are commonly used to assess marine environmental quality indicators (Palma et al., 2008).

The study of sediment compartment is fundamental since it allows us to estimate the quality of the prospected area. Such an approach should also benefit from a qualitative study leading to a diagnosis and an assessment of environmental quality. Assuming that the analytical process is correctly mastered, the researcher should then ask whether the observed temporal changes are real or due to a simple spatial variation.

To answer this question, it is first necessary to determine the spatial variability (Wu et al., 2008). This could be achieved by sampling sufficient replicates to provide reliable and statistically valid estimates of field data at a particular site or particular time, and/or discriminate differences between sampling sites and/or times. When replicates are collected in the field, the problem of spatial positioning of the samples is rarely taken into account (Birch et al., 2000; Liang and Wong, 2003; Lark et al., 2005; Morgan and Bull, 2007). Therefore, in this study, we seek to highlight the importance of spatial variability of environmental parameters at the scale of precision of the currently most commonly used positioning system (GNSS: Global Navigation Satellite System) to propose a sampling strategy for the monitoring of reference stations.

Due to their partly saline character substantially influenced by freshwater flows and their usually high ratio of sediment surface-area to water volume, coastal lagoons can be considered as very sensitive aquatic systems where benthic components and processes have an important regulatory function for the whole ecosystem (Viaroli et al., 2004). We consider that the main challenge, as already noted by Assinder et al. (2005) and Tolhurst and

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Chapman (2005), arises from the fact that the measured parameters exhibit spatial variations at scales near or even below the precision of the positioning system used. Furthermore, a real problem is posed by the analysis of collections of data that are positioned in space by systems having different accuracies.

With this objective in mind, we chose to study the Oualidia coastal lagoon (Morocco Atlantic coast), corresponding to an area where sediments are regularly exposed above sea level. The sampling strategy proposed here is allowing a high-precision geographical positioning of the sampling station.

2. Environmental parameters

Many coastal lagoons are mud and organic-enriched sedimentary systems. In this study, the investigated parameters (Fine Fraction Percentage, Total Organic Carbon, Inorganic Carbon and Total Sulphur) were chosen based on their importance in the assessment of coastal sites subject to strong anthropogenic pressures (Bryan and Langston, 1992; Bubbs and Lester, 1994; Daskalakis and O'Connor, 1995; Paez-Osuna et al., 1998; Magni et al., 2008; Molinaroli et al., 2009).

Together with sedimentation rate measurements (Doyle and Garrels, 1985), weight percent TOC is an important sediment determinant representing accumulation rates from sources such as domestic sewage, livestock waste discharges, industrial effluents, aquaculture activities, etc. It helps to explain the major accumulation of organic-enriched fine sediments in several lagoons (Rhoads, 1974; Rosenberg, 1978; Gray, 1979; Sorokin et al., 1996; Magni et al., 2005, 2008). It is also well known that the level of organic matter is related to the grain-size distribution of sediments (Tyson, 1995); in particular, higher organic matter contents tends to occur with increasing mud (clay) content due to the greater surface-area and higher number of complexing sites associated with the sedimentary particles (Buchanan and Longbottom, 1970; Mayer, 1994a,b; Tyson, 1995). With increased organic matter loading, sulphate reduction becomes the major metabolic pathway predominating over oxic (aerobic) respiration. Hence, total sulphur content increases with organic matter input (Hargrave et al., 2008) in marine areas. Carruesco (1989) demonstrated that carbonates in Oualidia lagoon are directly related to bioclastic marine sediments, so they may be considered as an indicator of marine influence.

The organic matter content in the sediments controls the structure and composition of macrobenthic assemblages in a lagoon system (Magni et al., 2008). The complex relationship between Fine Fraction Percentage, Total Organic Carbon, Inorganic Carbon and Total Sulphur clearly illustrates the importance of quantifying these parameters in each lagoon system (Florek and Rowe, 1983; Santschi et al., 1990; Fenchel et al., 1998) in order to evaluate the environmental quality of the studied area. As these parameters can be considered as the main keys for assessing the environmental quality of a coastal lagoon, their detailed analysis is crucial.

3. Spatial positioning

A Quality Assurance/Quality Control (QA/QC) procedure is required to obtain reliable data (Wells et al., 1993; Quarino et al., 1994; Chidi Ibe and Kullenberg, 1995; Quevauviller et al., 1995; Wagner, 1995; Teunissen, 1998; Tiberius, 1998; Taverniers et al., 2004). A satisfactory QA/QC procedure needs to be based on different control phases including technical aspects and the sedimentological materials used for testing. The first control phase includes several aspects such as the methods used to ensure representative sampling of environmental media, sample storage and standardization of processing procedures. Global Navigation Satellite System

(GNSS) is an important new technology for studying spatio-temporal behaviour and it has greatly aided the positioning of samples in the field. Surprisingly, almost no attention has been paid to the accuracy of sediment sampling positioning with GNSS and in particular GPS (Global Positioning System), one of the most commonly used and well known systems.

What is the spatial accuracy we could expect for a handled GPS? Accuracy is the degree of conformance between the measured position indicated by a GNSS receiver and the true position as compared with a fixed standard. GPS provides three-dimensional position locations from satellites to within about 100 m under normal operation, although the real accuracy is frequently as good as 20 m or even better (He et al., 2005; Luo et al., 2009). By operating in Differential mode, two DGPS units can give real-time locations accurate to less than a metre, and, with post-processing, to within a few millimetres (Nath et al., 2000).

In the scientific literature, GPS accuracy improved from 100 m before 1998 to 10–20 m after 1998 (Charpentier et al., 2005). The same order of accuracy is reported by Volpi Ghirardini et al. (2005) and better accuracy can be found in more recent studies: 10 m (Bell, 2009), 5 m (Lefebvre et al., 2009) and even better than ± 2 m (Dogan et al., 2009).

DGPS accuracy is considered to be about 2–3 m (Brooks and Mahnken, 2003; Gallagher et al., 2008; Ferentinos et al., 2010), 5 m (Brooks et al., 2003; Lefebvre et al., 2009; Schneider et al., 2010) to 10 m (Edwards, 2002; Figueiredo, 2009; Da Silva et al., 2009). It is very difficult to draw any conclusions about the real accuracy of the natural GPS and DGPS systems because no evidence of the accuracy is provided by the authors. Although positioning systems are very often cited as GPS, their reported accuracy is rather that of a DGPS system.

4. Study area and sampling

4.1. The Oualidia lagoon (Morocco)

The Oualidia lagoon (32°40'42"N–32°47'07"N and 8°52'30"W–9°02'50"W) is a shallow transitional system located on the Atlantic coast of Morocco (Fig. 1). The lagoon is 7 km long, on average 0.5 km wide, with an area of about 3 km², connected to the sea through two inlets (major and secondary inlets). The major inlets allow draining and filling of the lagoon by tidal flows, and also play an important role in feeding the inputs of tidal internal delta sediments of nearby sandy beaches (Zourarah, 2002).

The lagoon morphology is characterized by lateral channels, connected to a meandering main channel, with a mean depth of 2 m and a maximum depth during flood tides not exceeding 5 m (Bidet and Carruesco, 1982). This channel winds through a marsh 5.4 km long and about 0.4 km wide. Mud flats occupy mainly the low tide zone in the main channel and salt marshes are well developed throughout the lagoon, being invaded by halophytic vegetation consisting partly of samphire (Carruesco, 1989).

According to Carruesco (1989), sediments become finer from upstream to downstream, in correlation with a decrease in carbonate percentage. He explained that this intra-lagoon sediment distribution (coarse in the channel and the tidal delta and fine in the intertidal area) is influenced by tidal currents through inlets and also by major storms over the lagoon barrier. Tidal currents spread this material in the system, and their sorting leads to a distribution of fine-grained sediments in areas favourable for settling, such as in intertidal zones.

The Oualidia lagoon is typical of environments with a high anthropic pressure (sand extraction, salt marsh, mussel fishery, urban and tourism pressures) balanced by complex natural forcings (tides, swell, reworked sediment inputs, high biological productiv-

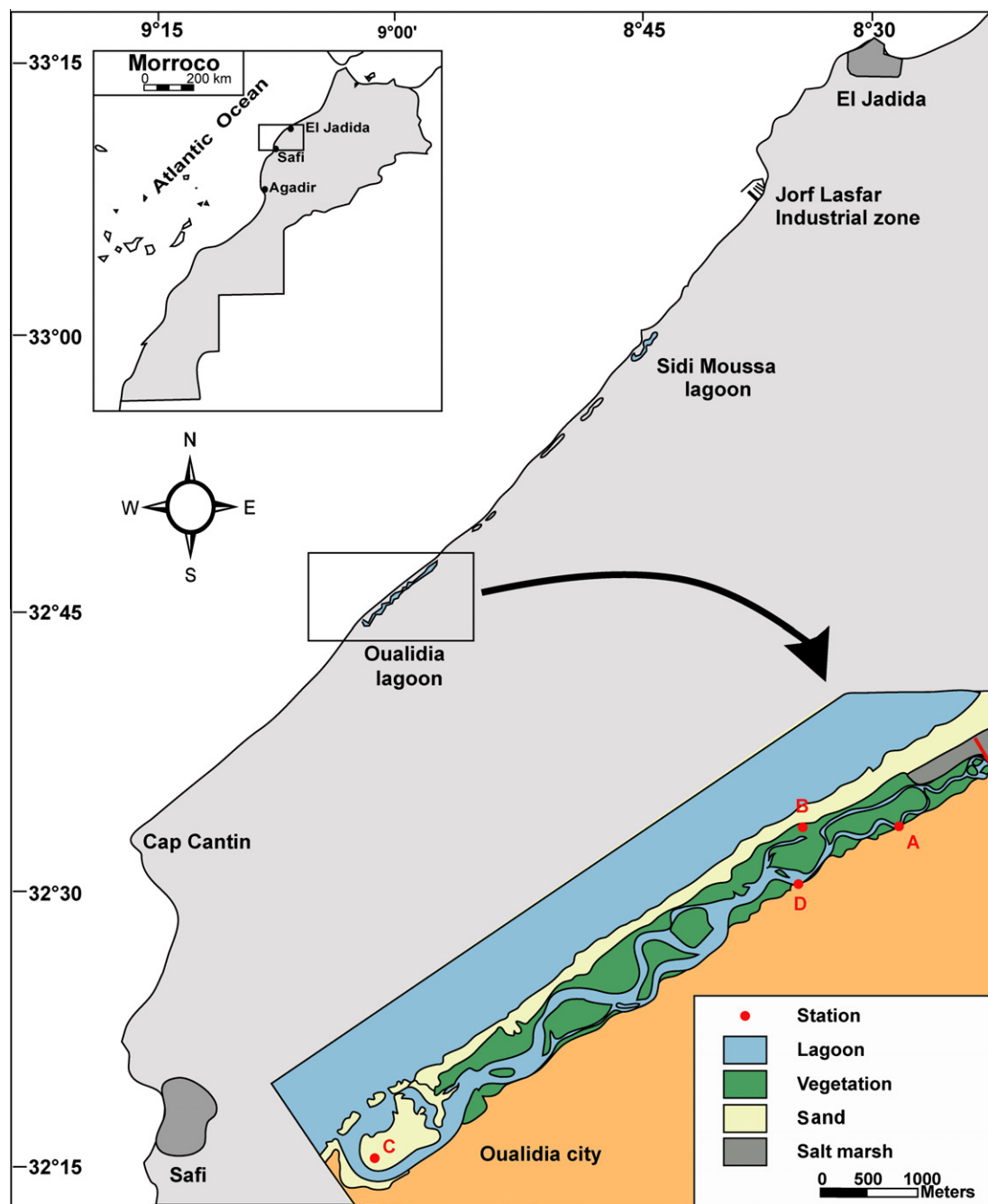


Fig. 1. Map of the studied coastal lagoon showing the sampling points.

ity, etc.). These interactions lead to an intense development of muddy areas with a high content in organic matter. The Oualidia lagoon is therefore a very good case study to illustrate the paradigm of geographical positioning accuracy.

4.2. Sampling sites

Four accessible sites were chosen in the Oualidia lagoon to represent a range of exposure conditions (Fig. 1). These sites were selected to characterize the different areas existing in the lagoon (Station A: mud flat near the channel, station B: vegetated marsh, station C: inlet sand bank and station D: mud flat).

All sites were sampled quantitatively on 24 October 2007 during low tide and under favourable weather conditions (no rain, no wind, etc.), so it may be considered that the quality of the sam-

ples is optimized. For all four of the stations sampled, the central points were positioned using a pocket GPS (Garmin III+) and a professional DGPS (Promark3). All other samples, situated around this central point, were positioned with a tape measure.

For this study, we adopted the sampling scheme proposed by MacKnight (1994). Each station consists of a circle with a diameter of approximately 30 m, having its central point fixed by geographical co-ordinates. We defined a physical circle of 30 m diameter for each station because this is sufficiently large to be identified without error using a non-differential GPS (ca. 15 m error) and it can be conveniently mapped at a scale of 1:10,000 to 1:25,000 (Volpi Ghiardini et al., 2005).

A cross-shaped sampling pattern was used, with samples being located at the centre of the cross and then on the four arms at regularly increasing distances, i.e. 50 cm, 1 m, 5 m and 15 m.

As a result, sixty-three samples of sediment (500 g) were collected with a plastic shovel. Sampled sediments corresponded to the topmost few cm of the surface layer. Samples were kept in air-sealed plastic bags and placed in a portable cooler at 4 °C, transported to the laboratory and stored in a freezer at –20 °C until subsequent analysis.

5. Methodology

5.1. Laboratory analysis

The reproducibility of any analytical system is of utmost importance, and is critical when working with small amounts of material to perform comparisons between sampling stations. For this reason, we used a very strict protocol which is described in detail below, thus allowing a better understanding of the accuracy and precision of the measurements.

First of all, sediment samples were freeze-dried. Then they were weighed (w_1) and wet sieved on a screen with a mesh of 1600 μm . The choice of this granulometric limit is due to the upper granulometric limit (2 mm) of the Beckman-Coulter LS 230 instrument. Both the finer (<1600 μm) and the coarser (>1600 μm) fractions were freeze-dried again and weighed (yielding w_2 and w_3 , respectively). The coarser fraction percentage was derived from w_3/w_1 . The finer fraction was then sub-sampled for granulometric and geochemical analysis.

The particle-size frequency distribution was determined in order to obtain the fine fraction percentage, i.e. the proportion (%) of the bulk sediment having a particle size <63 μm . In a first step, the samples were mixed with a dispersing agent (0.5% sodium hexametaphosphate) and left in deionised water for 2 h to disperse the clay particles. A Beckman-Coulter LS 230 laser diffractometer was used with a particle size analysis range of 0.04–2000 μm , divided into 116 fractions. At each class boundary, the particle size is 1.098 times coarser than the preceding class. All laser diffraction measurements were performed as suggested by the manufacturer (Beckman-Coulter, 1994). To prevent the formation of gas bubbles during the movement of suspended matter into the dispersion unit device, the stirrer velocity was set at 60–70 revolutions/s. The suspension was then pumped through a sample cell placed in the convergent laser beam, where forward scattered light impinged onto the 31 photosensitive sensor rings. Prior to each run, the detectors were aligned, the background measured and the sample dilution checked (to test that the sub-sample volume allowed a satisfactory analysis). Each run time was set at 60 s. The obscuration was adjusted to 40–50% for the measurements which included Polarization Intensity Differential Scattering (PIDS). All operations were controlled by a personal computer. For sediment suspensions, the small size of particles caused diffusion and reflection phenomena that needed to be taken into account using the Lorentz–Mie theory. The same optical model (i.e. with a complex refractive index of particles composed of a real and an imaginary part, the latter corresponding to absorption) must be selected for all sediment suspensions, to ensure consistent interpretations of scattering phenomenon, irrespective of whether particles occur as raw materials or as components of mixtures. This optical model allows us to extend the dimensional range of validity for the interpretation of scattering patterns, when using a PIDS system, to a particle size of about 0.04 μm . In this model, particles are assumed to be spherical, which allows granulometric distributions to be calculated in terms of volume percentage. According to previous studies (Bayle, 2004; ISO/TC24/SC 4 N 114), we adopt an optical model in which $n = 1.65 + 0.01i$. The model was selected by comparison with data from centrifugation sedimentation analysis based on Stokes' law (HORIBA Multisize Capa 400 particle-size analyser) in order to

avoid discrepancies that can occur as a function of particle shape (Novak and Thompson, 1985; Lohmander, 2000). All measurements were performed in triplicate and the results were analysed as volumetric distributions (Rawle, 1993).

To obtain samples for geochemical analysis, freeze-dried finer fractions were crushed and then homogenized. Total carbon contents and total sulphur content were measured by combustion in a LECO CS 300 carbon sulphur analyser. Three replicates of dried and homogenized sediment (50 mg) were analysed per sample following the established procedure. Samples were heated to 1600 °C and the amount of CO_2 and SO_2 was measured by infrared absorption.

For the analysis of Total Organic Carbon (TOC), sediment samples were acidified by HCl (12.5%) to remove carbonates, washed twice and dried on a hot plate at 50 °C. Inorganic carbon was obtained by difference. All inorganic carbon was assumed to be in the form of calcium carbonate.

Quality control was maintained by measuring LECO certified reference materials. Calibrations of the machine were performed regularly (2 calibrations between 6 analysis). The relative precision of repeated measurements of both samples and standards was in the range of 0.01–0.03 wt.%, the same as reported by Bechtel et al. (2002). Providing that appropriate standards (501 510, 501 501 and 501 503) and blanks are used, and that the subsamples analysed are fully representative of the whole set, the results should be both accurate and precise.

5.2. Principal component analysis

PCA, a multivariate analytical tool, was used to determine the distribution of the samples and to study the relationship of the measured parameters. PCA was conducted using the R computer software packages (R Development Core Team, 2008). PCA was performed on the whole standardized dataset (the mean and variance were set to 0 and 1, respectively) to minimize the effects of differences in measurement units or variance and to render the data dimensionless.

5.3. Positioning data acquisition

Currently most of sediment samples are positioned using a GPS device. However, when the used GPS mode is not mentioned, it is difficult to get the true spatial precision of the sampling scheme. In the present study, we compared two modes of the Global Positioning System (GPS) GNSS, i.e. natural GPS and differential GPS, to define the real spatial variability of environmental parameters (InC, TOC, F^2P and TS). The scale chosen (15 m) corresponds to the classical positioning precision obtained through the use of a good quality handheld GPS, which is the only accessible device in many developing countries lacking more precise systems.

To estimate the accuracy of our satellite positioning systems, we continuously recorded (frequency of 1 Hz), over a period of 2 h 30 min, the position of a fixed point situated inside the Safi University campus (Morocco). The data storage was carried out using two receivers, a GPS Garmin III+ and a DGPS Promark3 model from Thales using differential corrections by means of the Satellite Based Augmentation System (SBAS). The Garmin III+ receiver is less sophisticated and does not allow for the application of differential corrections. However, the Garmin receiver is simpler to use for non-specialist operators. On the contrary, the Promark3 is a professional receiver used for high-quality precision studies. It is therefore more complex to set up. In both cases, receivers were connected to a computer to store the geographical coordinates, which were then transmitted through an NMEA 183 (National Marine Electronics Association 183 standard) message (\$GPGGA).

The geodetic datums used were those of GPS/GNSS, i.e. WGS84 (World Geodetic System revised in 1984).

6. Results and discussion

6.1. Determination of GPS/DGPS precision

The mean position coordinates of the cloud of points (Fig. 2A) obtained during an interval of 150 min with the Garmin III+ receiver and the Thales receiver, respectively, are as follows: $X = -1031772.6$ m, $Y = 3806566.3$ m and $X = -1031771.5$ m, $Y = 3806563.9$ m (coordinates expressed with respect to the Google Mercator geodetic datum). The distance separating the two mean coordinates between the receivers is about 2.22 m, which corresponds to differences of 1.1 m and 2.4 m on the X and Y axes, respectively (Fig. 2B).

Hence, for the same acquisition rate (1 s), the two data sets show different spatial distributions. The set of data points acquired with the Promark3 (Fig. 2D) show an isotropic spatial distribution (80 cm in X and Y), while the data points obtained with the Garmin III+ (Fig. 2C) show an elliptical spatial organization elongated North–South (major axis = 13 m and minor axis = 10 m).

Inside the cloud of points, the spatial distribution is quite different between the two receivers. For the Garmin III+, points are separated by a distance of about 1.6 m on the X axis and 1.9 m on the Y axis. These values correspond to the best precision that can be achieved by the Garmin III+ receiver. On the contrary, the Promark3 receiver is able to attain a precision of 2 mm on both X and Y axes. This difference is due to the coding of coordinates in the NMEA183 \$GPGGA message sent by the two receivers. The Garmin III+ receiver outputs have a precision of a thousandth of a minute, while the Promark3 outputs have a precision of a millionth of a minute.

Precision is significantly different between the two receivers. The Promark3 receiver allows a precision of 30 cm on both axes, whereas the precision of the Garmin III+ receiver is 2.9 m on the X axis and 4.6 m on the Y axis.

The accuracy of the two receivers is also very different (Fig. 3). Whatever the axis (X or Y), the accuracy of the Promark3 receiver is

higher than the Garmin III+ receiver. It is noteworthy that the Garmin III+ receiver X coordinates are shifted to the bottom, when the Y coordinates are shifted to the top, which causes the elliptical shape of the cloud of points given by the Garmin III+ receiver.

It appears that the two receivers are of different quality. While the professional Promark3 receiver can be considered as precise and accurate, this cannot be said of the Garmin III+ receiver. When a long acquisition time is considered, there is no great difference between the two datasets. However, this would not be the case for a real sampling operation on the field. Under such conditions, when randomly taking just a single value, the difference can be around 20 m.

In the framework of a temporal study, geographical coordinates appear unsuitable when acquired with a GPS receiver such as Garmin III+, so it is necessary to use a GPS receiver equipped with an Augmentation System using satellites or terrestrial radio beacons. To avoid undermining the reliability of sample locations when returning to the reference station, it is necessary to analyse replicates that are sampled in a circle of few meters radius centred on the expected reference station location. It should be noted that a professional GPS receiver needs to be operated by a person properly trained in the use of the device. This type of accurate professional device also offers a wide choice of settings, which, if they are not perfectly mastered, can lead to invalid results without the user being aware.

6.2. Environmental variability

6.2.1. Statistical results

The PCA analysis was performed on dataset containing four variables (F^2P , TOC, InC and TS) and 63 individuals. First two principal components (PCs) were responsible for 94% of the total variance. PC1 account for 84% and PC2 for 10% on the total variance (Fig. 4).

The variables factor map (Fig. 4A) exhibits two sets of variables. The first one is composed of F^2P , TOC and TS parameters with correlation coefficient ranging from 0.64 to 0.79. The second set contain only the InC parameter. The two sets are highly negatively correlated ($R^2 = -0.86$) which explain the height level of variance

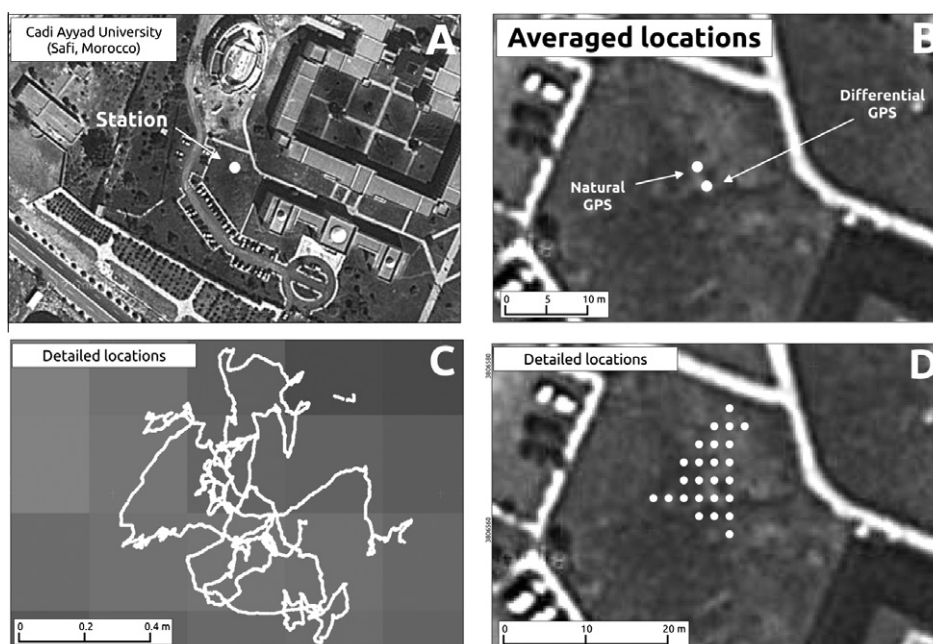


Fig. 2. Maps of the GPS coordinates obtained with a Garmin III+ and a Promark3 receiver. (A) geographical location of the station; (B) geographical position given by the two devices (averaged positions); (C) spatial distribution of the position points given by Garmin III+; (D) spatial distribution of the position points given by Promark3.

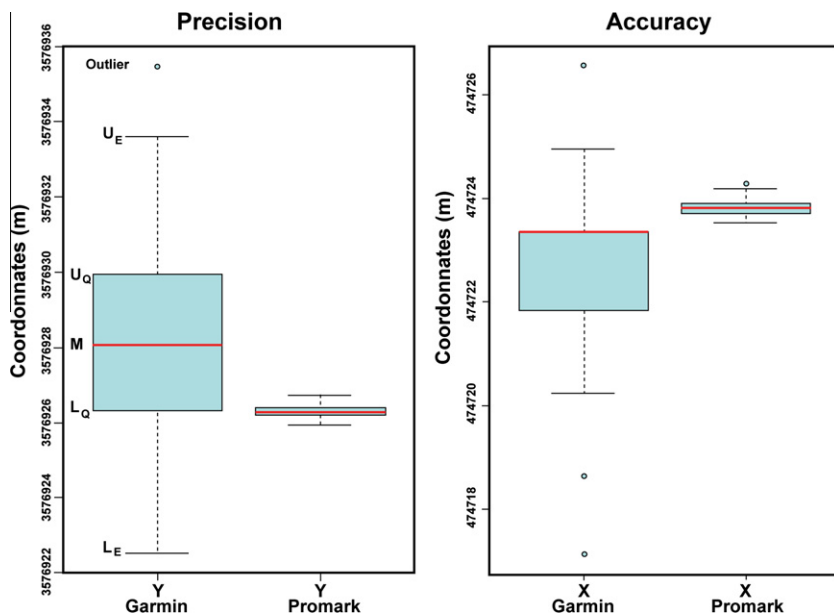


Fig. 3. Box and whiskers plot illustrating the difference in repeatability of positioning data obtained by the Garmin III+ and Promark3 receivers along the X and Y axes. LE: Lower Extreme; LQ: Lower Quartile; M: Median; UQ: Upper Quartile; UE: Upper Extreme.

explained by PC1. In the frame of a lagoon environment, the InC set represents the marine pole whereas F²P, TOC and TS set represent the continental pole.

As shown by the score plot, sediment samples had obvious spatial clustering (Fig. 4B). Four clusters are distinguished, one for each station. They exhibit a spatial organization between the two poles. The station C is close to the marine pole and the station B cluster to the continental pole. The station A cluster is stretch between the marine pole and the continental pole. The station D cluster is in mid position slightly closer to the continental pole. In the course of this alignment, each station cluster areas successively increase. Then, individuals spreading is variable depending on the considered station.

The station C is the smallest cluster, i.e. all the samples are gathered in a very small area. This station is spatially homogeneous.

The station A shows an elongated cluster with an internal organization depending on the distance to the central station point. This latter point and the nearest half meter sample points are concentrated near the marine pole. Samples at 1 and 5 m progressively move away to the continental pole. Sample at 15 m distance form a disconnected cluster close to the station D.

The station D cluster has a rounded shape with the sampled central point at its centre. However, no internal organization depending on the distance of the central point, can be highlighted.

The station B cluster is very spread. The central sampling point is at the left outer edge. It does not belongs to any of the fourth samples groups collected at a common distance (0.5, 1, 5 and 15 m). These latter does not show any logical organization within the cluster. Then, the station B exhibit a strong random heterogeneity.

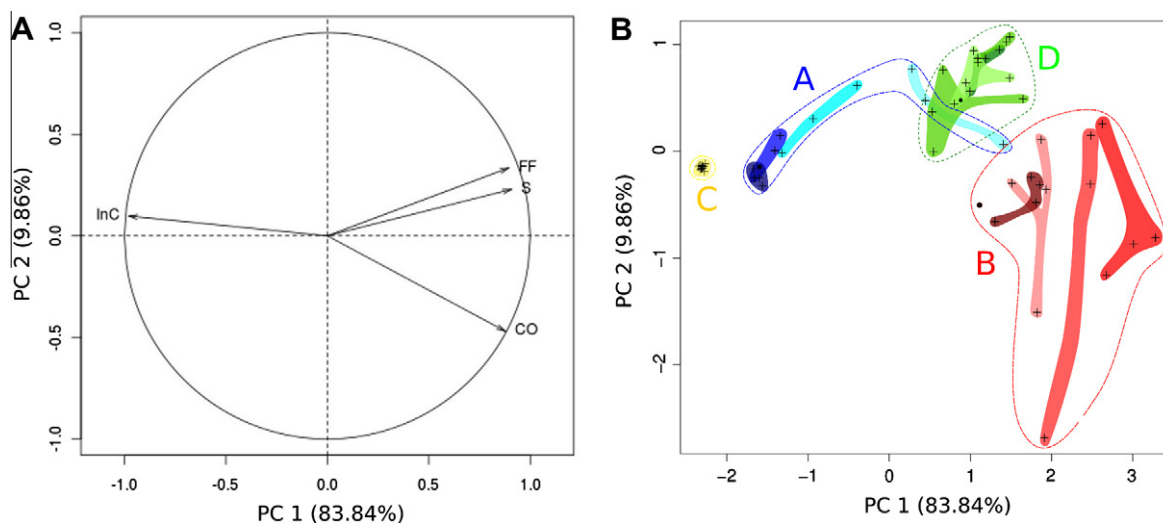


Fig. 4. (A) Principal component variable loading plot. (B) Cluster of the sampling stations by PCA. Plot of principal component 1 versus 2. (A–D) are for studied stations. For each cluster, colour intensity represents the distance from the central point. Darker is the colour, closer are the individuals from the central point. For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.

Table 1

Ratio factor (max value/min value) defined for each parameter at each station.

Station	Spatial homogeneity			
	More		Less	
A	InC x 2	TS X 6	TOC X 10	F ² P X 10
B	TOC x 2	F ² P X 3	TS X 5	InC X 10
C	F ² P =	InC =	TOC =	TS =
D	TS X 1.4	InC X 1.4	TOC X 1.6	F ² P X 3

The spatial variability was also investigated through a ratio factor. It has been defined as the ratio between the maximum and minimum value for each parameter at each station (Table 1).

6.2.2. Spatial variability inside station

6.2.2.1. Mud flat near the channel (station A). The study area (Fig. 1) corresponds to an intertidal mud flat, without any development of vegetation. The central point of the reference station is located immediately above the central main channel of the lagoon, more precisely, near a convex part of the channel.

Station A was not completely sampled on the four axes away from the central location (Fig. 5). Sampling towards the west was

abandoned due to the strong hydrodynamic regime occurring in the central channel.

At the centre of the station, and at distances of up to 1 m away, sediments are characterised by a first granulometric mode around 360 µm (medium sand) and another mode around 800 µm (coarse sand). At 5 m from the central point, the sediments exhibit a trimodal distribution, with the appearance of fine silt (mode of 6–10 µm). 15 m from the central point, sediments are finer (mode of 120 µm). The granulometric curve shows a fine particle tail characteristic of a positively skewed distribution.

The multi-modal granulometry of this station is due to sediment movements in this complex hydrodynamic setting (seaward inputs into the lagoon even at some distance for the inlet, multiple twisting channels), and the “pollution” of the site by local inputs of reworked sediments. This complexity arises from the different types of sediment grains (of different density, shape, etc.) which show rapidly changing proportions.

Station A shows a pattern in the spatial distribution of the studied parameters. Gradients are observed away from the central area in the three studied directions. The fine fraction percentage increases towards the mud flat with increasing distance away from the highly dynamic channel area. The increase in mineral carbonate is due to the proximity of the channel, and hence is related to the marine influence.

The TOC increases along with the fine fraction percentage, and hence towards the mud flat. The farthest point of this station is situated 15 m to the East of the central point. This extreme sampling

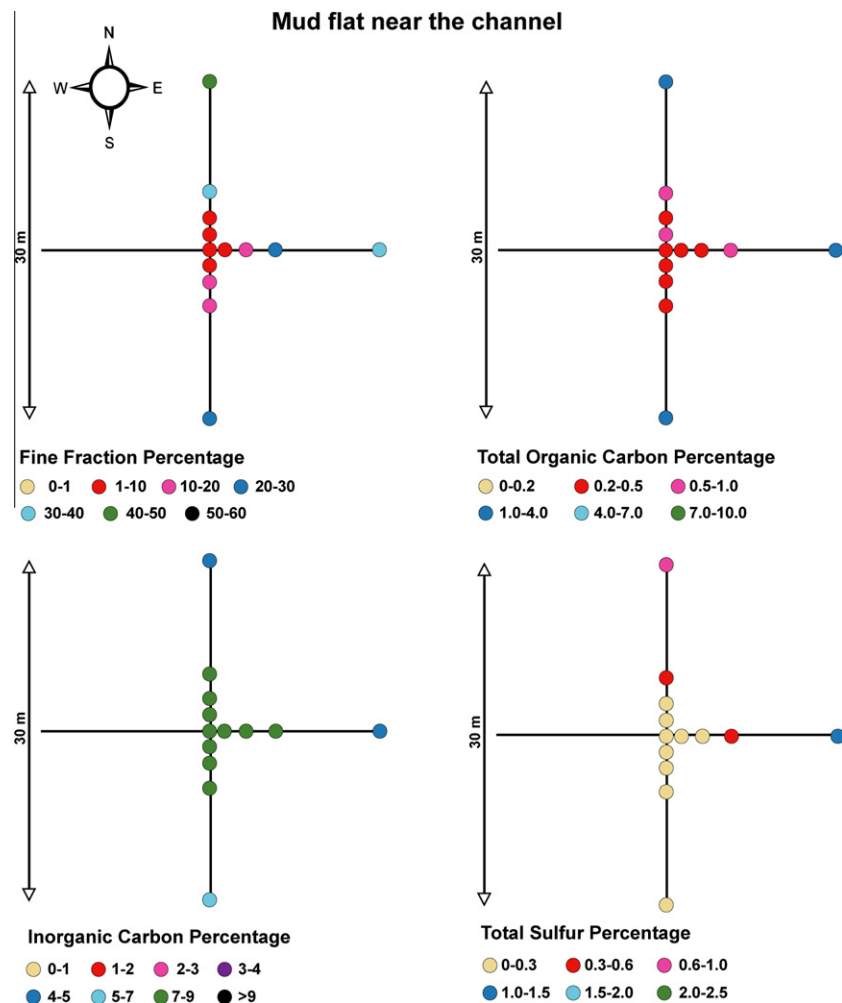


Fig. 5. Spatial distribution of the four studied parameters around central point of station A.

point yields the highest TOC percentage (4%). At this location, the granulometric fraction coarser than 1600 μm is mostly composed of the macro debris of continental plants. This reflects the low-energy hydrodynamic regime as well as the proximity of continental inputs.

Total sulphur content rises in conjunction with the TOC, and consequently with the fine fraction percentage, and hence also towards the mud flat. In a marine environment, this trend highlights the development of sulphate reduction associated with increased TOC percentage.

The central part is homogeneous up to at least 50 cm from its centre (fine fraction percentage) and more generally for the other parameters at a distance of 1 m. Taking this spatial heterogeneity into account, we can rank the degree of homogeneity of the studied parameters (Table 1).

6.2.2.2. Marsh (station B). Station B is situated in a marsh zone with development of vegetation. The highly complex vegetation distribution made it impossible to choose a sufficiently homogeneous station at a distance of 30 m. By default, the central point of the station was placed in a homogeneous area of smaller dimension without any vegetation (Fig. 6).

Station B was sampled along its four axes away from the central point. Only the BW4 point could not be sampled.

The characteristic sediments of station B show the same grain-size distribution, with two main modes. The coarser mode (around 750 μm) is the same for all the samples. It is composed of plant and shell debris. The finer mode differs according to the plant coverage at the sample location, varying from 90–120 μm in areas without vegetation to 105–250 μm for areas with a vegetation cover. For samples characteristic of areas lacking vegetation, the coarser fraction is characterized by debris made up almost exclusively of marine gastropods. For the other samples, where the vegetation is variably present, plant debris can represent up to 100% of the coarser fraction.

Station B shows the highest spatial variability. The central point, lacking vegetation, is characterized by a low fine fraction content (17.6%), the maximum InC (3.3%), the lowest TOC (4.0%) and a total sulphur percentage equal to the mean of the area (1.6%).

The area heterogeneity is variable according to the considered parameters. It starts at the distance of 50 cm as regard for the fine fraction, 1 m for the InC and the TOC and finally 5 m for the TS. For the fine fraction and InC, the variability is high and the distribution of values does not follow a spatially coherent pattern.

For TOC, the variability is moderate (Table 1) with a higher frequency of values at 4–5%, which can be considered as background noise. Higher percentages (>7.5%) are considered as random. The spatial variability of these three parameters could be associated with (1) the vegetation cover (in particular for TOC), (2) the pres-

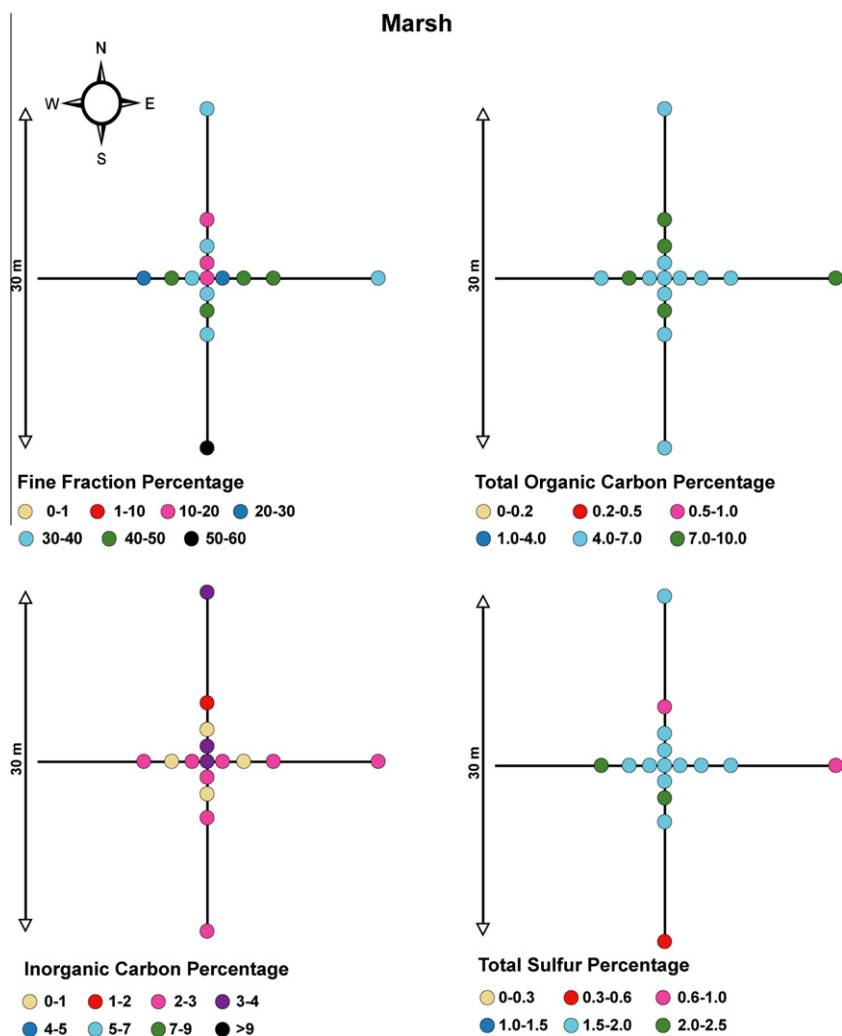


Fig. 6. Spatial distribution of the four studied parameters around central point of station B.

ence of small draining channels on the marsh (in particular for the fine fraction percent), (3) bathymetric variations (in particular for InC).

For Total Sulphur, the sampled area shows a random heterogeneity at 5 m from the central point. To account for the spatial variability of the sulphur contents, we can evoke the reasons cited above. However, the salinity factor should also be considered. The areas with the densest vegetation cover, which also have the highest elevation, can only rarely be reached by the tide. Therefore, the salinity in such areas will be low, and associated with limited sulphate reduction, while the percentage of sulphur will be lower even though the TOC percentage is high (because of the vegetation).

For the station representative of the marsh, spatial heterogeneity is apparent over small distances. Only sulphur percentage shows homogeneity at 1 m from the central point. The fine fraction percent is not homogeneous at 50 cm. This station is characteristic of an area where it is very difficult to establish a temporal tracking. Based on the spatial heterogeneity observed at this station, we can rank the variability of the studied parameters (Table 1).

6.2.2.3. Inlet sand bank (station C). Station C is situated in a sandy area, in front of the inlets which allow communication between the coastal lagoon and the Atlantic ocean. It is characteristic of a high-energy hydrodynamic environment always submerged at high tide.

Station C was completely sampled along the four axes from the central point, providing a total of 17 samples (Fig. 7).

Sediments are composed of shell debris. The granulometry at this station is highly homogeneous whatever the distance of the sample from the central point of the station. The mode is about 280 μm , which leads us to classify these sediments as medium sands.

Station C is homogeneous at all scales for all the parameters (F^2P : 0.5%; TOC: 0.08% on average; InC: 11.5%; TS: 0.24% on average). This station is characteristic of an area where temporal studies are easy to implement. There is a high probability that this homogeneity is also temporal. Based on the spatial heterogeneity observed at this station, it is possible to rank the studied parameters (Table 1).

6.2.2.4. Mud flat (station D). Station D is located in the middle part of the lagoon, more precisely in the intertidal area. Compared to station B, which is located at the same distance from the inlet, vegetation is not present. Moreover, station D differs from station A by its distance from the centre of the channel. It was sampled along its four axes from the central point, thus providing a total of 17 samples (Fig. 8).

The main granulometric mode ranges from 100 to 250 μm , without any coherent spatial pattern. The coarser mode (750 μm) is composed of plant and shell debris in similar amounts for all the samples. This station is subject to marine influence, as

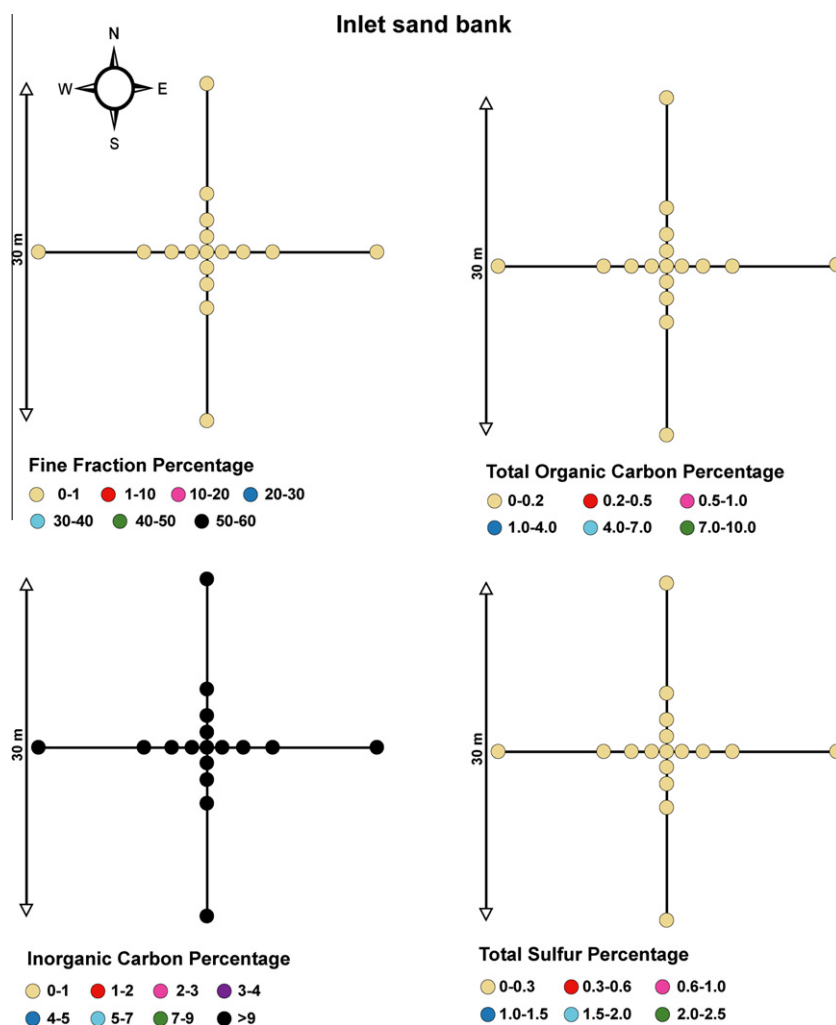


Fig. 7. Spatial distribution of the four studied parameters around central point of station C.

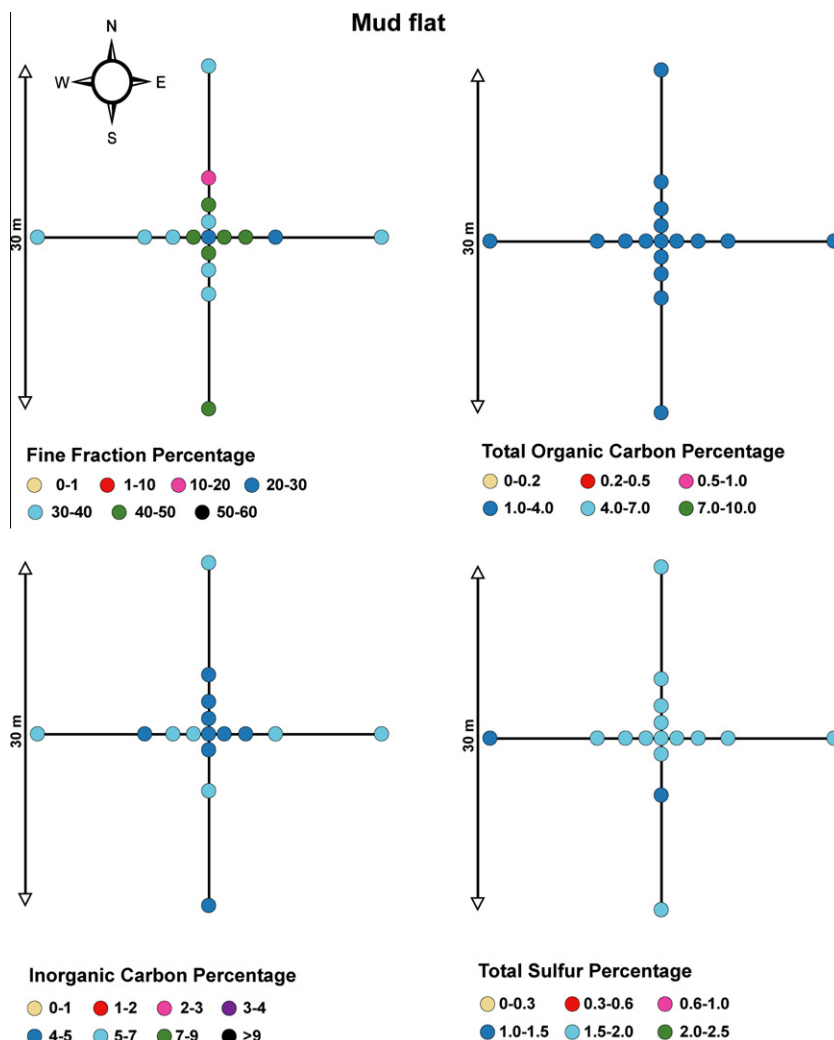


Fig. 8. Spatial distribution of the four studied parameters around central point of station D.

highlighted by a mineral carbon content systematically higher than 4%.

Station D is characterized by a fine fraction percent with a high spatial heterogeneity. This latter characteristic is surprising due to the environmental context (no vegetation). This could be explained by the presence of small drainage channels on the mud flat, which could be responsible for enhanced hydrodynamic energy during the ebb tide. On the contrary, InC, TOC and TS have a relatively homogeneous spatial variability (InC: 5.0%; TOC: 2.3%; TS: 1.6%). Even considering this low variability, no spatial organization can be observed. Based on their degree of spatial heterogeneity, the variability of the studied parameters can be ranked (Table 1).

6.2.3. Environmental synthesis

Inside the Oualidia coastal lagoon, four different environmental areas were studied which show contrasted spatial variability.

The shell sand bank located near the lagoon inlet (station C) exhibits a high homogeneity of the deposits, whatever the studied parameter, over the entire surface area (15 m in all directions). This sand bank can be considered as being built up under high-energy hydrodynamic conditions and the resulting sediment can be assumed to be homogeneous at distances greater than those used during the sampling.

The mud flat without vegetation and far from the channel (station D) shows a low degree of spatial variation for the three studied geochemical parameters (InC, TOC and TS). In the framework of a

site used for time-tracking of the environment, this station can be considered as homogeneous over its entire area (30 m diameter). On the contrary, the fine fraction percent shows a random spatial heterogeneity outwards from a distance 50 cm.

The mud flat near the channel (station A) is characterized by a coherent spatial distribution of the four studied parameters. There is always a gradient from the central point (near the channel) to the extremities of the three studied directions. The central part can be considered as homogeneous within a zone extending from 0.5 cm to 1 m from the central point.

The marsh with discontinuous vegetation cover (station B) corresponds to an area showing the highest spatial heterogeneity whatever the studied parameters. At most, the homogeneous zone is 1 m in diameter (TOC, InC and TS), whereas the fine fraction percent becomes heterogeneous at distances greater than 0.5 cm.

Two environments (i.e. the sand shell bank and the mud flat near the channel) can be easily monitored for the purpose of temporal tracking. They are the most subject to marine influence and can be considered as being homogeneous over the distances investigated by the DGPS receiver (1 m). The marsh (station D) and the mud flat far from the channel (station B), representing a greater continental influence, are more complex to use for environmental temporal tracking. In the case of the mud flat, only the fine fraction percent gives rise to problems in temporal tracking, but it is nevertheless a fundamental environment parameter.

For the marsh area, all the parameters are spatially heterogeneous, but again, the fine fraction percent shows the highest variability. In the case of temporal tracking, these two latter environments require extra efforts for a single sampling period, in order to carry out a spatial variability study, so that more samples would have to be collected.

6.2.4. Synthesis of parameter variability

Four parameters were studied (F^2P , TOC, InC and TS) in environments showing high variability of measured values (by a ratio factor ranging from 6 to 10, for fine fraction percent at station A, InC at station B, TOC at station A and TS at station C). Geochemical parameters show the smallest spatial variability ($\text{InC} < \text{TOC} < \text{TS}$). Even for environments where the spatial variability is strong (station B), some homogeneity is always found at least 50 cm from the central point. For the three other stations, this zone of homogeneity extends to 1 m (station A), 5 m (stations A and D) and 15 m (stations C and D). The fine fraction percent exhibits the opposite behaviour, showing a high degree of random spatial variability in 2 of the 4 studied environments (stations B and D), that is to say, without any spatial auto-correlation at the studied distances.

In conclusion, some parameters (InC, TOC, TS) show a sufficiently low spatial variability in most of the studied environments to allow temporal tracking through GNSS positioning systems (professional devices) without multiplying the number of samples. By contrast, the fine fraction percent (F^2P) shows a high spatial variability over small distances. Hence, the spatial variability of this parameter needs to be studied in the framework of temporal study during each period of sampling, which leads to a multiplication in the number of samples.

7. Conclusion

To optimize an environmental study, it is necessary to define the reference station central point location by means of a DGPS with metric precision. The complexity of this type of device should be taken into account before its implementation in the field, to avoid introducing more uncertainties into the positioning measurements than those classically associated with the use of a pocket handled GPS. The points adjacent to the central point of the sampling array should then be placed geographically with a tape measure.

Regardless of the parameters studied, it is risky to characterize an environment by taking a single sample when the spatial variability is unknown. The number and arrangement of replicates depends on both the environment and the parameters studied.

We propose a two-step protocol to optimize the procedure to be carried out as part of a quality assurance approach:

- (1) The first step is to establish the scale of the spatial variability specific to the environment studied and the chosen parameters. We recommend adopting the metric scale appropriate to the accuracy of the DGPS, involving a sampling scheme with a central point and four directions (orthogonal), with a sample collected at 50 cm, 1 m and 2 m from the central point, that is to say 13 samples to be analysed from a given monitoring station.
- (2) The second step is to choose a sampling strategy for the temporal tracking of the station. The degree of heterogeneity can be determined according to the results of step 1.

If the area is:

- homogeneous over a range of 2 m, one sample will suffice for temporal tracking;

- homogeneous over a range of 1 m, five samples must be sampled (one at the central point and 4 at 50 cm, according to a cross-shaped pattern);
- homogeneous over a range of 50 cm, nine samples (one at the central point, 4 at 50 cm and 4 at 1 m) are needed for satisfactory temporal tracking;
- heterogeneous beyond 50 cm, the spatial variability is greater than the accuracy of DGPS, and it is unrealistic to consider a valid reference station.

This protocol tested here and is proposed for intertidal areas where accurate positioning is available. In permanently submerged areas, the high accuracy provided by GPS devices is illusory because of (1) the movement of the boat, (2) the accuracy of the offset measurement (distances between the sampling device and the GPS antenna), so the proposed methodology cannot be applied at the same geographical scale.

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References

- Assinder, D.J., Mudge, S.M., Russell, A.T., 2005. Micro- and meso-scale variability in sediment activity concentrations: the implications for dose assessment. *Radioprotection* 40 (1), S569–S573.
- Bayle, S., 2004. Etude de suspensions hétérogènes pour le contrôle d'un procédé papetier Dimensions et propriétés optiques de la matière: des paramètres discriminants. PhD Thesis, Université Montpellier II, France.
- Bechtel, A., Butuzova, L., Turchanina, O., 2002. Thermochemical and geochemical characteristics of sulphur coals. *Fuel Processing Technology* 77–78, 45–52.
- Beckman-Coulter, 1994. Coulter LS Series: Product Manual. Beckman Coulter, Miami, FL, 345pp.
- Bell, J.J., 2009. Hitching a ride on a hermit crabs home: movement of gastropod shells inhabited by hermit crabs. *Estuarine, Coastal and Shelf Science* 85, 173–178.
- Bidet, J.C., Carruesco, C., 1982. Étude sédimentologique de la lagune de Oualidia (Maroc). In: Lasserre, P. et al. (Eds.), *Coastal Lagoons: Proceedings of the International Symposium on Coastal Lagoons*, 8–14 September 1981, Bordeaux, France. *Oceanologica Acta* 4, pp. 29–37.
- Birch, G.F., Taylor, S.E., Matthai, C., 2000. Small-scale spatial and temporal variance in the concentration of heavy metals in aquatic sediments: a review and some new concepts. *Environmental Pollution* 113 (3), 357–370.
- Birch, G.F., Taylor, S.E., Matthai, C., 2001. Small-scale spatial and temporal variance in the concentration of heavy metals in aquatic sediments: a review and some new concepts. *Environmental Pollution* 113, 357–372.
- Brooks, K.M., Mahnken, C.V.W., 2003. Interactions of Atlantic salmon in the Pacific Northwest environment III. Accumulation of zinc and copper. *Fisheries Research* 62, 295–305.
- Brooks, G.R., Doyle, L.J., Suthard, B.C., Locker, S.D., Hine, A.C., 2003. Facies architecture of the mixed carbonate/siliciclastic inner continental shelf of west-central Florida: implications for Holocene barrier development. *Marine Geology* 200, 325–349.
- Bryan, G.W., Langston, W.J., 1992. Bioavailability, accumulation and effects of heavy metals in sediments with special reference to United Kingdom estuaries: a review. *Environmental Pollution* 76, 89–131.
- Bubb, J.M., Lester, J.N., 1994. Anthropogenic heavy metal inputs to lowland river systems, a case study. *The River Stour UK. Water, Air, & Soil Pollution* 78, 279–296.
- Buchanan, J.B., Longbottom, M.R., 1970. The determination of organic matter in marine muds. The effect of the presence of coral and the routine determination of protein. *Journal of Experimental Marine Biology and Ecology* 5, 158–169.
- Carruesco, C., 1989. Genèse et évolution de trois lagunes du littoral atlantique depuis l'holocène: Oualidia, Moulay Bousselham (Maroc) et Arcachon (France), Thèse d'État, Université Bordeaux-1, France.
- Charpentier, A., Grillas, P., Lescuyer, F., Coulet, E., Auby, I., 2005. Spatio-temporal dynamics of a *Zostera noltii* dominated community over a period of fluctuating salinity in a shallow lagoon, Southern France. *Estuarine, Coastal and Shelf Science* 64, 307–315.

- Chidi Ibe, A., Kullenberg, G., 1995. Quality Assurance/Quality Control (QA/QC) regime in marine pollution monitoring programmes: the GIPME perspective. *Marine Pollution Bulletin* 31, 209–213.
- Da Silva, A.C., Potma, K., Weissenberger, J.A.W., Whalen, M.T., Mabilbe, C., Boulvain, F., 2009. Magnetic susceptibility evolution and sedimentary environments on carbonate platform sediments and atolls, comparison of the Frasnian from Belgium and from Alberta. *Sedimentary Geology* 214, 3–18.
- Daskalakis, K.D., O'Connor, T.P., 1995. Normalization and elemental sediment contamination in the coastal United States. *Environmental Science & Technology* 29 (2), 470–477.
- Dogan, O.K., Akyurek, Z., Bekioglu, M., 2009. Identification and mapping of submerged plants in a shallow lake using quickbird satellite data. *Journal of Environmental Management* 90, 2138–2143.
- Doyle, L.J., Garrels, M., 1985. What does percent organic carbon in sediment measure? *Geo-Marine Letters* 5, 51–53.
- Edwards, B.D., 2002. Variations in sediment texture on the northern Monterey Bay National Marine Sanctuary continental shelf. *Marine Geology* 181, 83–100.
- Fenchel, T., King, G.M., Blackburn, T.H., 1998. Bacterial Biogeochemistry: The Ecophysiology of Mineral Cycling. Academic Press, San Diego, 307pp.
- Ferentinos, G., Papatheodorou, G., Geraga, M., Iatrou, M., Fakiris, E., Christodoulou, D., Dimitriou, E., Koutsikopoulos, C., 2010. Fjord water circulation patterns and dysoxic/anoxic conditions in a Mediterranean semi-enclosed embayment in the Amvrakikos Gulf, Greece. *Estuarine, Coastal and Shelf Science* 88, 473–481.
- Figueiredo, J., 2009. Sedimentological and Stratigraphic Evolution of a Muddy Submarine Slope Succession, Karoo Basin, South Africa. Unpubl. PhD Thesis, University of Liverpool, 202pp.
- Florek, R.J., Rowe, G.T., 1983. Oxygen consumption and dissolved inorganic nutrient production in marine coastal and shelf sediments of the Middle Atlantic Bight. *Internationale Revue der Gesamten Hydrobiologie* 68, 73–112.
- Frontier, S., 1983. L'échantillonnage de la diversité spécifique. In: *Stratégie d'échantillonnage en écologie*, Frontier et Masson édité, Paris (Coll. d'Ecologie), XVIII, 494pp.
- Gallagher, F.J., Pechmann, I., Bogden, J.D., Grabosky, J., Weis, P., 2008. Soil metal concentrations and vegetative assemblage structure in an urban brownfield. *Environmental Pollution* 153, 351–361.
- Gray, J.S., 1979. Pollution-induced changes in populations. *Philosophical Transactions of the Royal Society B* 286, 545–561.
- Hargrave, B.T., Holmer, M., Newcombe, C.P., 2008. Towards a classification of organic enrichment in marine sediments based on biogeochemical indicators. *Marine Pollution Bulletin* 56, 810–824.
- He, Y., Yu, H., Fang, H., 2005. Study on improving GPS measurement accuracy. In: *Instrumentation and Measurement Technology Conference 2005. Proceedings of the IEEE* 2, pp. 1476–1479.
- ISO/TC24/SC 4 N 114. Particle size analysis, Guide to Laser Diffraction Methods.
- Lark, R.M., Bellamy, P.H., Rawlins, B.G., 2005. Spatial-temporal variability of some metal concentrations in the soil of eastern England, and implications for soil monitoring. *Geoderma* 133 (3–4), 363–379.
- Lefebvre, A., Thompson, C.E.L., Collins, K.J., Amos, C.L., 2009. Use of a high-resolution profiling sonar and a towed video camera to map a *Zostera marina* bed, Solent, UK. *Estuarine, Coastal and Shelf Science* 82, 323–334.
- Liang, Y., Wong, M.H., 2003. Spatial and temporal organic and heavy metal pollution at Mai Po Marshes Nature Reserve, Hong Kong. *Chemosphere* 52, 1658–1674.
- Lohmander, S., 2000. Aspect ratio of pigment particles determined by different methods. *Nordic Pulp & Paper Research Journal* 3, 221–230.
- Luo, X., Mayer, M., Heck, B., 2009. On the issues of probability distribution of GPS carrier phase observations. *General Assembly of the European Geosciences Union 2009, Wien, Österreich*, Geophysical Research Abstracts 11, 4815.
- MacKnight, S., 1994. Selection of bottom sediment sampling stations. In: *Mudroch, A., MacKnight, S. (Eds.), Handbook of Techniques for Aquatic Sediments Sampling*, second ed. CRC Press Inc.,
- Magni, P., Micheletti, S., Casu, D., Floris, A., Giordani, G., Petrov, A.N., De Falco, G., Castelli, A., 2005. Relationships between chemical characteristics of sediments and macrofaunal communities in the Cabras lagoon (Western Mediterranean, Italy). *Hydrobiologia* 550, 105–119.
- Magni, P., Rajagopal, S., van der Velde, G., Fenzi, G., Kassenberg, J., Vizzini, S., Mazzola, A., Giordani, G., 2008. Sediment features, macrozoobenthic assemblages and trophic relationships (d13C and d15N analysis) following a dystrophic event with anoxia and sulphide development in the Santa Giusta lagoon (western Sardinia, Italy). *Marine Pollution Bulletin* 57, 125–136.
- Mayer, L.M., 1994a. Surface area control of organic carbon accumulation in continental shelf sediments. *Geochimica et Cosmochimica Acta* 58, 1271–1284.
- Mayer, L.M., 1994b. Relationships between mineral surfaces and organic carbon concentrations in soils and sediments. *Chemical Geology* 114, 347–363.
- Molinarioli, E., Guerzoni, S., Sarretta, A., Masiol, M., Pistolato, M., 2009. Thirty-year changes (1970–2000) in bathymetry and sediment texture recorded in the Lagoon of Venice sub-basins, Italy. *Marine Geology* 258, 115–125.
- Morgan, R.M., Bull, P.A., 2007. The use of grain size distribution analysis of sediments and soils in forensic enquiry. *Science and Justice* 47 (3), 125–135.
- Nath, S.S., Bolte, J.P., Ross, L.G., Aguilar-Manjarez, J., 2000. Applications of Geographical Information Systems (GIS) for spatial decision support in aquaculture. *Aquacultural Engineering* 23, 233–278.
- Novak, J.W., Thompson, J.R., 1985. Extending the use of particle sizing instrumentation to calculate particle shape factors. *Powder Technology* 45, 159–167.
- Paez-Osuna, F., Bojorquez-Leyva, H., Green-Ruiz, C., 1998. Total carbohydrates: organic carbon in lagoon sediments as an indicator of organic effluents from agriculture and sugar-cane industry. *Environmental Pollution* 102, 321–326.
- Palma, E.D., Matano, R.P., Piola, A.R., 2008. A numerical study of the Southwestern Atlantic Shelf circulation: stratified ocean response to local and offshore forcing. *Journal of Geophysical Research* 113, C11010.
- Quarino, L., Wilson, R., Ferrara, J., Baum, H., Shaler, R.C., 1994. Quality assurance in a forensic biology laboratory. *Forensic Science International* 68 (1), 17–28.
- Quevauviller, P., Maier, E.A., Griepink, B., 1995. Quality assurance for environmental analysis. In: *Quevauviller, P., Maier, E.L., Griepink, B. (Eds.), Quality Assurance for Environmental Analysis*. Elsevier, Amsterdam, Nederland, pp. 1–25.
- R Development Core Team (2008). R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, URL <http://www.R-project.org>.
- Rawle, A., 1993. Basic Principles of Particle Size Analysis, MRK034, Malvern Instruments.
- Rhoads, D.C., 1974. Organism-sediment relations on the muddy sea floor. *Oceanography and Marine Biology: An Annual Review* 12, 263–300.
- Rosenberg, D.M., 1978. Practical sampling of freshwater macrozoobenthos: a bibliography of useful texts, reviews and recent papers. Technical report. Fisheries and Marine Service, Canada 790, 1–15.
- Santschi, P.H., Bollhalder, S., Zingg, S., Luck, A., Farrenkothent, K., 1990. The self-cleaning capacity of surface waters after radioactive fallout. Evidence from European waters after chernobyl, 1986–1988. *Environmental Science & Technology* 24 (4), 519–527.
- Schneider, M., Romero, P.M., Hase, F., Blumenstock, T., Cuevas, E., Ramos, R., 2010. Continuous quality assessment of atmospheric water vapour measurement techniques: FTIR, Cimel, MFRSR, GPS, and Vaisala RS92. *Atmospheric Measurement Techniques* 3, 323–338.
- Sorokin, Y.I., Sorokin, Y.P., Giovanardi, O., Venezia, D.L., 1996. Study of the ecosystem of the Lagoon of Venice, with emphasis on anthropogenic impact. *Marine Ecology Progress Series* 141, 247–261.
- Taverniers, I., De Loose, M., Van Bockstaele, E., 2004. Trends in quality in the analytical laboratory. II. Analytical method validation and quality assurance. *Trends in Analytical Chemistry* 23 (8), 535–552.
- Teunissen, P.J.G., 1998. Quality control and GPS. In: *Teunissen, P.J.G., Kleusberg, A. (Eds.), GPS for Geodesy*, second ed. Springer, Verlag (Chapter 7).
- Tiberius, C.C.J.M., 1998. Quality control in positioning. *Hydrographic Journal* 90, 3–8.
- Tolhurst, T.J., Chapman, M.G., 2005. Spatial and temporal variation in the sediment properties of an intertidal mangrove forest: implications for sampling. *Journal of Experimental Marine Biology and Ecology* 317, 213–223.
- Tyson, R.V., 1995. Sedimentary Organic Matter. Organic Facies and Palynofacies. Chapman and Hall, London, 615pp.
- Viaroli, P., Bartoli, M., Giordani, G., Magni, P., Welsh, D.T., 2004. Biogeochemical indicators as tools for assessing sediment quality/vulnerability in transitional aquatic ecosystems. *Aquatic Conservation: Marine and Freshwater Ecosystems* 14, S19–S29.
- Volpi Ghirardini, A., Arizzi Novelli, A., Tagliapietra, D., 2005. Sediment toxicity assessment in the Lagoon of Venice (Italy) using *Paracentrotus lividus* (Echinodermata: Echinoidea) fertilization and embryo bioassays. *Environment International* 31, 1065–1077.
- Wagner, G., 1995. Basic approaches and methods for quality assurance and quality control in sample collection and storage for environmental monitoring. *The Science of Total Environment* 176, 63–71.
- Wells, D.E., Cofino, W.P., Quevauviller, P., Griepink, B., 1993. Quality assurance of information in marine monitoring: a holistic approach. *Marine Pollution Bulletin* 26, 368–375.
- Wu, R.S.S., Chan, A.K.Y., Richardson, B., Au, D.W.T., Fong, J.K.H., Lam, P.K.S., Giesy, J.P., 2008. Measuring and monitoring POPs: a critique. interdisciplinary studies on environmental chemistry, biological responses to chemical pollutants, Murakami, Y., Nakayama, K., Kitamura, S.I., Iwata, H., Tanabe S., (Eds.), pp. 1–6.
- Zourarah, B., 2002. Les processus côtiers actuels et leur impact sur l'environnement côtier des Doukkala. Thèse d'Etat, Université Chouaib Doukkali, Faculté des Sciences, El Jadida, Maroc, 376pp.